

Scott Okuma

Technical Program Leadership | Hardware/Software System Integration

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Summary

Systems architect and engineering leader with 14+ years across spacecraft, embedded computing, and RF systems. Built Umbra's flight-software organization, managed 12 direct reports, and collaborated with approximately 20 electrical and mechanical engineers to deliver integrated spacecraft capabilities. Experience spans technical project management, hardware/software interfaces, development processes, test planning, and deployment across three spacecraft blocks successfully operating in orbit. Combines hands-on electrical engineering with leadership of multidisciplinary development and integration.

Professional Experience

Umbra Space

Spacecraft Systems Architect | Goleta, CA | March 2025 to Present

Director of Spacecraft Software | Santa Barbara, CA | August 2021 to March 2025

Flight Software Lead | Santa Barbara, CA | June 2019 to September 2021

Selected accomplishments across Umbra roles

- Built Umbra's flight-software department from the ground up; hired and managed 12 direct reports and established architecture, development processes, test infrastructure, and delivery practices.
- Served as project manager for two engineering projects, alongside spacecraft architecture and firmware leadership responsibilities.
- Drove spacecraft platform architecture with approximately 20 electrical and mechanical engineers, coordinating hardware/software interfaces across flight computers, guidance, navigation and control (GNC), payload, radio, avionics, and ground-command systems.
- Led the transition from bare-metal flight software to embedded Linux, spanning firmware, FPGA, Linux kernel development, and GNC model integration across the computing platform.
- Delivered flight software across three spacecraft blocks successfully deployed to orbit, with no mission losses attributable to flight software.
- Established nightly CI/CD and software test infrastructure to support integration and delivery; developed environmental test plans covering execution, teardown, and pre-/post-test functional validation for thermal-vacuum, shock, and vibration verification.
- Reviewed processor schematics and PCB layouts to verify pin assignments and required hardware/software interfaces; made component-selection and architecture decisions for radiation-tolerant systems, including FRAM nonvolatile memory.
- Architected flight-proven remote-update capabilities across four onboard computers, with staged updates, checksum validation, redundant boot images, and automatic rollback to support continued operation without physical access.
- Architected dual-flight-computer redundancy and fault recovery, including watchdog supervision, automatic failover, sensor/actuator resets, and low-power sun-pointing safe mode.

RF, Hardware, and Test Engineering

Electrical/Firmware Engineer

Raytheon - Goleta, CA | September 2017 to June 2019

- Designed digital firmware for RF system controllers using VHDL/Verilog and Vivado; customized Linux kernels and developed embedded systems with PetaLinux and C/C++.
- Conducted hardware checkout, schematic analysis, and DDR3 implementation for electronic modules, using oscilloscopes, spectrum analyzers, and network analyzers.

Electrical Engineer

NAWCWD - Point Mugu, CA | January 2015 to September 2017

- Developed software and performed RF hardware repair, layout, radar calibration, and test planning for radar systems.
- Created a DRFM jammer using an FPGA, RF transceiver, and user interface; programmed in Python, C++, and MATLAB for signal processing and simulation.

Electrical/RF Engineer

Computer Sciences Corporation - Point Mugu, CA | January 2012 to January 2015

- Developed automated test procedures for GaN RF amplifiers using LabVIEW and GPIB; built custom test fixtures and collaborated with technicians to troubleshoot systems.
- Verified equipment thermal limits using environmental chambers and data logging; analyzed test results using Octave.

University Projects

CSUN CubeSat1 - Satellite Communications Team Member

Jet Propulsion Laboratory / CSU Northridge | January 2014 to December 2014

Developed radio communication test procedures for electrostatic-sensitive devices and bit-error-rate, link-budget, and data-volume analyses for mission planning.

NOAA Radio Weather Receiver Team Member

CSU Northridge | February 2014 to June 2014

Designed an RF amplifier for a superheterodyne receiver and developed a software-defined radio using MATLAB/Simulink.

Integration and Program Capabilities

Program delivery: Technical project management; cross-functional engineering coordination; team development; hiring and mentoring; development processes; CI/CD and delivery practices.

System integration: Hardware/software architecture; avionics interfaces; schematic and PCB review; ARM and Zynq platforms; FPGA/processor interfaces; embedded Linux; GNC model integration.

Test and verification: Environmental test planning; thermal-vacuum, shock, and vibration verification; automated RF testing; LabVIEW/GPIB; oscilloscopes; spectrum and network analyzers.

RF and software: RF system controllers; S-band radios; software-defined radio; C/C++; Python; VHDL/Verilog; MATLAB/Simulink; PetaLinux; Yocto; Vivado/Vitis; Git/GitLab.

Education and Training

Bachelor of Science in Electrical Engineering | California State University Northridge | 2012-2014

Xilinx Embedded Design with PetaLinux Tools | November 2018